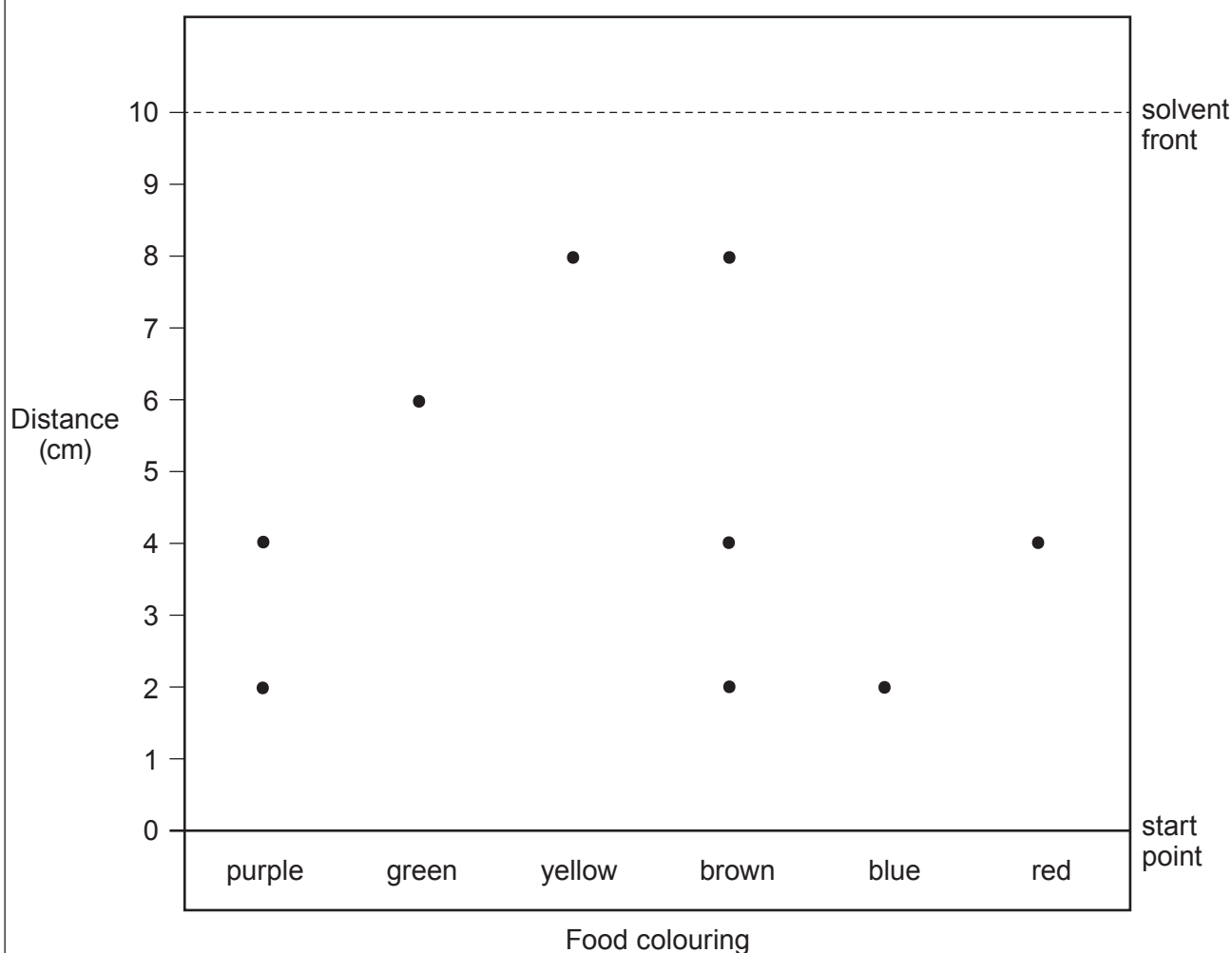


2. Analytical scientists work in a wide range of different industries and agencies.

- (a) The Food Standards Agency (FSA) funded a project to analyse the colours used in food and drinks.

The diagram below shows a chromatogram of the dyes present in different food colourings.



- (i) It was thought that the dye in the blue colouring would not be present in other colourings.
Explain whether you agree.

[2]

.....

.....



(ii) Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the R_f value for the dye in the **red** colouring.

[3]

$R_f =$

- (b) Analytical scientists also identify unknown compounds.
There were three bottles, each containing a solution of one of the following compounds.

copper(II) sulfate

lithium carbonate

sodium chloride

- (i) State which of these compounds would produce a yellow flame in a flame test. [1]

- (ii) I. State which of these compounds would produce bubbles of gas when added to hydrochloric acid. [1]

- II. The gas turns limewater milky. Name this gas. [1]

- (iii) 1 cm³ of silver nitrate solution is added to three different test tubes, each containing a solution of one of the compounds in the box above.

- I. State which of these compounds will give a white precipitate. [1]

- II. Complete the risk assessment below for silver nitrate. [2]

Hazard	Risk	Control measure
silver nitrate is irritant	could cause eye damage if splashes in the eyes when	
		
		
		

